

Mixtures & Solutions

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A **mixture** combines materials that keep their own properties and can be separated again (like trail mix). A **solution** is a special mixture that looks the same throughout, with one material **dissolved** in another (like salt water). Ways to separate a mixture: **sieve, filter, magnet, or evaporation.**

Round 1 — Retrieve It

1. What is a mixture?

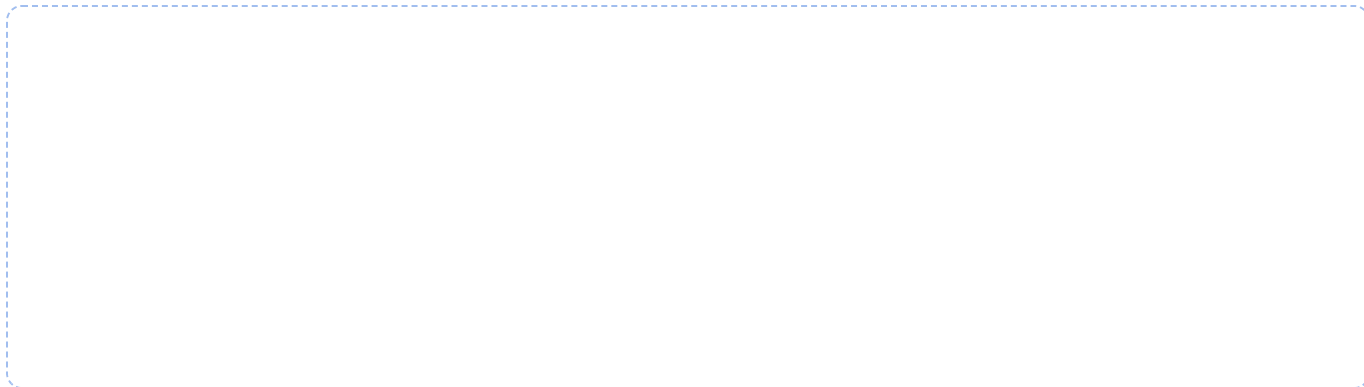
2. What is a solution? Give one example.

3. Name one way to separate a mixture.

4. When you make a solution, what happens to the material that is dissolved?

Round 2 — Sketch It

Draw two cups. In one cup, show a mixture where you can still see the separate parts. In the other, show a solution that looks the same throughout. Label each cup.



Mixtures & Solutions — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. You have a mixture of sand and iron filings. Which tool would separate them, and why does it work?

2. You have salt water and want to get the salt back. What method would you use, and what happens to the water?

3. You want to separate small pebbles from sand. Would a sieve or a magnet work better? Explain.

Round 4 — Reason from Evidence

A student mixes sugar into water until they can no longer see it. Is this a mixture or a solution? Use evidence to explain how you know, and whether the sugar is still there.

Mixtures & Solutions — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how a solution is different from other mixtures. Use the words dissolved and looks the same throughout.

C**Connect**

Connect a separation method (sieve, filter, magnet, or evaporation) to a real-life job or chore where someone would use it.

E**Extend**

Design a mixture that contains at least three materials. Plan a step-by-step way to separate all of them, and name the tool for each step.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Mixtures & Solutions

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Materials combined together that keep their own properties and can be separated again (such as trail mix).
2. A mixture that looks the same throughout, with one material dissolved in another. Example: salt water (or sugar water).
3. Any: sieve, filter, magnet, or evaporation.
4. It dissolves and spreads out evenly, so you can no longer see it, but it is still there.

Sketch It — look for

One cup shows separate, visible parts (a mixture). The other cup looks the same all the way through with no visible pieces (a solution). Both cups are labeled correctly.

Apply It

1. A magnet; iron filings are attracted to the magnet and the sand is not, so the magnet pulls the iron out.
2. Evaporation; the water turns into a gas and leaves, and the salt stays behind.
3. A sieve works better; its holes let the smaller sand fall through while the larger pebbles stay on top. A magnet would not work because neither is magnetic.

Reason from Evidence — look for

It is a solution because the sugar dissolved and the liquid looks the same throughout. The sugar is still there (you could taste it or get it back by evaporating the water); it did not disappear.

ACE — Extend look-for

Plan lists three materials and a workable order of steps, matching each material to a sensible tool (magnet for magnetic items, sieve for large pieces, filter for solids in liquid, evaporation for a dissolved solid).

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.